

Syllabus: Surrogate Modeling for Process Prediction and Optimization

Build validated process predictors, learn optimization formulations and model structures, then select and revalidate operating conditions.

Course design and notes: Sunwoo Kim · Graduate entry · Updated 2026-10-09

Course purpose

Start by learning a process response from data: define inputs and outputs, fit a surrogate, and check its predictions on independent conditions. Then examine physical consistency and learn how a trained function can be represented inside an optimization problem. Finally, choose operating conditions and revalidate them with the reference process.

The course follows **prediction** → **physical consistency** → **optimization formulations and model structure** → **operating decisions**. It covers both formulating an already-trained model and designing a model structure for an intended optimization use. Prediction error, physical consistency, formulation agreement, and decision quality are evaluated separately.

Designed by Sunwoo Kim as an independent, self-paced course for chemical engineers entering graduate study. Original explanations, worked readings, and runnable examples accompany verified research sources.

Audience and prerequisites

Basic Python, derivatives, matrix operations, linear algebra, and material balances are expected. Prior neural-network or mathematical-programming research experience is not required. LP/MILP, convexity, and KKT conditions are introduced when needed.

Suggested weekly workload: 90 minutes of theory, 60–90 minutes of guided coding, and 2–3 hours of reading or exercises. These are study recommendations, not scheduled classes.

Learning outcomes

By the end of the course, learners should be able to:

1. Define a process prediction task: inputs, outputs, fixed context, states, units, data splits, and valid domain.
2. Fit steady-state and dynamic surrogates, and evaluate component errors and trajectory errors using independent data and prescribed future inputs.
3. Compare unconstrained, soft-physics, and hard linear-equality models, separating prediction accuracy from physical consistency.
4. Define operating variables, an objective, and constraints; formulate a frozen ReLU model as a valid MILP and check agreement with its forward pass.

5. Select model structure and formulation for the intended use, including the conditions under which ReLU ICNN/PICNN epigraphs permit an equivalent LP problem.
6. Solve a surrogate-based operating problem and report reference-model feasibility, economic performance, solver termination, gaps, and time separately.

Eight-week sequence

Week	Topic and scope	Required output
1	Process Prediction and Surrogate Modeling Basics. Inputs, outputs, fixed context, units, operating domain, reference data, a regression baseline, and independent prediction checks.	A steady-state CSTR data contract, fitted regression baseline, and held-out prediction report.
2	Learning Multicomponent Steady-State Responses with MLPs. Three-component balances, scaling, MLP forward/backpropagation, training, component errors, and concentration consistency.	A steady-state MLP evaluated in physical units, with a hand-checked gradient update.
3	Predicting Dynamic Responses and Rollout. State-transition MLPs, initial states, future operating inputs, one-step prediction, and full-trajectory rollout. RNN/LSTM and encoder–decoder extensions are optional.	A comparison of one-step and trajectory errors under prescribed future inputs.
4	Learning with Physical Constraints. Residual losses, automatic differentiation, hard linear-equality projection, KKT derivation, rank assumptions, and remaining violations.	An MLP/soft-physics/hard-equality comparison separating accuracy from physical consistency.
5	From a Predictor to an Optimization Formulation. Operating variables versus fixed context, objectives and constraints, frozen ReLU graphs, binary variables, Big-M, valid bounds, and scaling.	A hand-built MILP representation checked against the frozen network forward pass.
6	Building Formulations We Can Solve. Bound tightening, scaling, OMLT, relaxations, solver status, gaps, computation time, and optional short-horizon extensions.	An operating optimization solve with documented bounds, termination, gaps, timing, and forward rechecks.
7	Designing Model Structure for Optimization. ICNN/PICNN, convexity conditions, ReLU/linear epigraphs, fixed context, inequality direction, and equivalence counterexamples.	An ICNN and valid LP on a known convex example, with explicit applicability conditions.
8	Selecting and Revalidating Operating Conditions. Decision-point errors, extrapolation, physical feasibility, economic performance, reference-model checks, and targeted additional data.	A reproducible prediction-to-optimization report separating errors, violations, decision quality, and computation.

Weeks 1–3 establish prediction. Week 4 examines physical consistency. Weeks 5–6 formulate and solve problems with trained models; Week 7 designs model structure for optimization. Week 8 integrates the full workflow. Operating decisions are introduced briefly as the course destination in Week 1.

The syllabus and **Week 1–Week 2** core materials are available. Weeks 3–8 remain planned. The existing dynamic material in the Week 2 appendix is an optional Week 3 preview; later operating-point examples are also optional.

Reading assignments

Week	Reading guidance
1	Week 1 reading companion; Boyd & Vandenberghe, affine mappings (optional).
2	Week 2 reading companion; selected PyTorch model, autograd, and training tutorials.
3	Course state-transition derivation and the supplied dynamic preview; Sutskever et al., Sections 1–2 (optional).
4	Raissi et al., residual-loss formulation; Chen et al., Section 3 and selected process cases.
5	Boyd & Vandenberghe, optimization problems (selected); Anderson et al., ReLU formulations.
6	OMLT paper and documentation; selected stronger-formulation material from Anderson et al.
7	Amos et al., Section 3 and Supplement B.
8	Revisit the model and formulation assumptions, validation evidence, and reference results from Weeks 1–7.

Start with the supplied **Week 1 reading companion** and **Week 2 reading companion**, each available as English/Korean PDFs. External readings use selected sections. RNN/LSTM, encoder–decoder, DeepONet, advanced formulation proofs, and longer-horizon examples are optional extensions.

Core sources

- **Convex optimization background:** S. Boyd and L. Vandenberghe, *Convex Optimization* (2004). Read selected parts of Chapters 2–4 for convex sets, convex functions, epigraphs, and optimization problems; return to Chapter 5 for duality and KKT. **Official book and downloads** · **EE364a slides**.
- **Neural-network implementation:** PyTorch tutorials on **model construction**, **autograd**, and **training**.
- **ICNN/PICNN:** B. Amos, L. Xu, and J. Z. Kolter, *Input Convex Neural Networks*, ICML (2017). Section 3 gives architecture conditions; Supplement B gives LP inference for ReLU/linear units. **Paper** · **PDF** · **Supplement**.
- **ReLU formulations:** R. Anderson, J. Huchette, W. Ma, C. Tjandraatmadja, and J. P. Vielma, *Strong mixed-integer programming formulations for trained neural networks*, Mathematical Programming (2020). The public manuscript was first submitted in 2018 and revised in 2020. **Public manuscript**.
- **Embedding software:** F. Ceccon, J. Jalving, J. Haddad, A. Thebelt, C. Tsay, C. D. Laird, and R. Misener, *OMLT: Optimization & Machine Learning Toolkit*, JMLR 23(349), 1–8 (2022). **Paper and PDF** · **Code** · **Documentation**.
- **PINN:** M. Raissi, P. Perdikaris, and G. E. Karniadakis, *Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations*, Journal of Computational Physics 378, 686–707 (2019). **Published paper**. The related 2017 Part I manuscript is a freely available introduction; it is not the identical final journal version. **2017 Part I**.
- **KKT-hPINN:** H. Chen, G. E. Constante Flores, and C. Li, *Physics-informed neural networks with hard linear equality constraints*, Computers & Chemical Engineering 189, 108764 (2024). **Published paper** · **Public manuscript** · **Code and datasets**.
- **Sequence-learning background:** I. Sutskever, O. Vinyals, and Q. V. Le, *Sequence to Sequence Learning with Neural Networks*, NeurIPS (2014). Its application is translation; our process-trajectory exercises are separate educational examples. **Paper**.

Optional extensions: **DeepONet, Nature Machine Intelligence (2021)**, for input-function to output-function mappings; **KKT-Hardnet public manuscript (2025), published in C&CE (2026)**, for nonlinear equality and inequality projection.

Process examples and software

Week 1 uses a small ideal isothermal CSTR, reference-generated data, and a quadratic regression baseline. Week 2 extends to $A \rightarrow B \rightarrow C$ and a steady-state MLP. Week 3 uses state transitions and rollout as the common dynamic example. These teaching models run without Aspen or plant data; they do not establish the computational cost or fidelity of an industrial simulator.

Week 4 uses residual losses and consistent linear equalities; public KKT-hPINN datasets are a separate reproduction exercise. Weeks 5–6 reuse a frozen ReLU predictor for optimization. Week 7 first checks convexity and LP equivalence on a separate known convex target; an ICNN approximation does not prove that a CSTR or every flowsheet output is convex.

Week 1 requires Python, NumPy, and Matplotlib. Week 2 adds PyTorch. Later weeks introduce Pyomo/OMLT and a suitable solver. The Week 1 script runs prediction by default; `--extensions` retains the earlier operating-selection and dynamic examples for later study.

What “exact” and “hard” mean here

- **Exact embedding:** a formulation represents the frozen learned function over the stated domain. It does not remove approximation error relative to the physical process. MILP status also depends on the rest of the objective and constraints.
- **LP reformulation:** ReLU/linear ICNN structure, fixed PICNN context, valid nonnegative propagation, and appropriate epigraph usage matter. Arbitrary output equalities, reversed inequalities, smooth activations, or additional nonlinear constraints can invalidate an LP claim.
- **Hard constraints:** KKT-hPINN enforces the specified consistent linear equalities under its matrix assumptions, up to numerical arithmetic. It does not automatically enforce positivity, nonlinear thermodynamics, stability, or every operating limit.
- **Optimization with physics-informed models:** training losses do not determine the algebraic class of the frozen predictor. A ReLU backbone followed by a fixed affine equality projection remains piecewise affine; a smooth or nonlinear projection can change the required formulation.

Exercises and final project

Week 1 submits a prediction specification, fitted baseline, and held-out evaluation. Week 2 submits a steady-state MLP, checked gradient update, and component/consistency diagnostics. Later weeks add verified elements.

Document the domain, data split, model structure, frozen weights, scaling, formulation assumptions, solver termination, and reference process. Report prediction error, physical consistency, formulation agreement, and decision quality separately. Compare economics with the reference model. Identify grid or local-solver benchmarks without claiming certified global process optimality.

Assessment covers mathematical correctness, reproducibility, decision evidence, and accurate reporting of unresolved error or feasibility. Include failed operating points and incomplete solves.